Lecture No.2 Truth Tables

Truth Tables

Truth Tables for:

1. **$\sim p \wedge q$**
2. **$\sim p \wedge (q \vee \sim r)$**
3. **$(p \vee q) \wedge \sim (p \wedge q)$**

Truth table for the statement form $\sim p \wedge q$

p	q	~p	~p ∧ q
T	T	F	F
T	F	F	F
F	T	T	T
F	F	T	F

Truth table for $\sim p \wedge (q \vee \sim r)$

p	q	r	~ r	q ∨ ~ r	~ p	~ p ∧ (q ∨ ~ r)
T	T	T	F	T	F	F
T	T	F	T	T	F	F
T	F	T	F	F	F	F
T	F	F	T	T	F	F
F	T	T	F	T	T	T
F	T	F	T	T	T	T
F	F	T	F	F	T	F
F	F	F	T	T	T	T

Truth table for $(p \vee q) \wedge \sim (p \wedge q)$

p	q	p∨q	p∧q	~ (p∧q)	(p∨q) ∧ ~ (p∧q)
T	T	T	T	F	F
T	F	T	F	T	T
F	T	T	F	T	T
F	F	F	F	T	F

Double Negative Property $\sim(\sim p) \equiv p$

p	~p	~(~p)
T	F	T
F	T	F

Example
"It is not true that I am not happy"
Solution:
Let p = "I am happy"
then ~ **p** = "I am not happy"
and ~(~ **p)** = "It is not true that I am not happy"
Since ~ (~**p) ≡ p**
Hence the given statement is equivalent to:
"I am happy"

~ (p∧q) and ~p ∧ ~q are not logically equivalent

p	q	~p	~q	p∧q	~(p∧q)	~p ∧ ~q
T	T	F	F	T	F	F
T	F	F	T	F	T	F
F	T	T	F	F	T	F
F	F	T	T	F	T	T

Different truth values in row 2 and row 3

DE MORGAN'S LAWS:

1) The negation of an **and** statement is logically equivalent to the **or** statement in which each component is negated.
Symbolically ~(p ∧ q) ≡ ~p ∨ ~q.
2) The negation of an **or** statement is logically equivalent to the **and** statement in which each component is negated.
Symbolically: ~(p ∨ q) ≡ ~p ∧ ~q.

~(p ∨ q) ≡ ~p ∧ ~q

p	q	~p	~q	p ∨ q	~(p ∨ q)	~p ∧ ~q
T	T	F	F	T	F	F
T	F	F	T	T	F	F
F	T	T	F	T	F	F
F	F	T	T	F	T	T

Same truth values

Application:

Give negations for each of the following statements:

a. The fan is slow **or** it is very hot.

b. Akram is unfit **and** Saleem is injured.

Solution

a. The fan is **not** slow **and** it is **not** very hot.

b. Akram is **not** unfit **or** Saleem is **not** injured.

INEQUALITIES AND DEMORGAN'S LAWS:

Use DeMorgan's Laws to write the negation of

$-1 < x \leq 4$

for some particular real no. x

$-1 < x \leq 4$ means $x > -1$ **and** $x \leq 4$

By DeMorgan's Law, the negation is:

$x > -1$ or $x \leq 4$Which is equivalent to: $x \leq -1$ **or** $x > 4$

EXERCISE:

1. $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
2. Are the statements $(p \wedge q) \vee r$ and $p \wedge$□ $(q \vee r)$ logically equivalent?

TAUTOLOGY:

A tautology is a statement form that is always true regardless of the truth values of the statement variables.

A tautology is represented by the symbol "**T**"..

EXAMPLE: The statement form $p \vee \sim p$ is tautology

p	~p	p ∨ ~p
T	F	T
F	T	T

$p \vee \sim p \equiv t$

CONTRADICTION:

A contradiction is a statement form that is always false regardless of the truth values of the statement variables.

A contradiction is represented by the symbol "c".

So if we have to prove that a given statement form is **CONTRADICTION** we will make the truth table for the statement form and if in the column of the given statement form all the entries are F ,then we say that statement form is contradiction.

EXAMPLE:

The statement form $p \wedge \sim p$ is a contradiction.

p	$\sim p$	$p \wedge \sim p$
T	F	F
F	T	F

Since in the last column in the truth table we have F in all the entries so is a contradiction $p \wedge \sim p \equiv c$

REMARKS:

- Most statements are neither tautologies nor contradictions.
- The negation of a tautology is a contradiction and vice versa.
- In common usage we sometimes say that two statements are contradictory.

By this we mean that their conjunction is a contradiction: they cannot both be true.

LOGICAL EQUIVALENCE INVOLVING TAUTOLOGY

1. Show that $p \wedge t \equiv p$

p	t	$p \wedge t$
T	T	T
F	T	F

Since in the above table the entries in the first and last columns are identical so we have the corresponding statement forms are Logically Equivalent that is

$$p \wedge t \equiv p$$

LOGICAL EQUIVALENCE INVOLVING CONTRADICTION

Show that $p \wedge c \equiv c$

p	c	$p \wedge c$
T	F	F
F	F	F

Same truth values in the indicated columns so $p \wedge c \equiv c$

EXERCISE:

Use truth table to show that $(p \wedge q) \vee (\sim p \vee (p \wedge \sim q))$ is a tautology.

SOLUTION:

Since we have to show that the given statement form is Tautology so the column of the above proposition in the truth table will have all entries as T.

As clear from the table below

p	Q	$p \wedge q$	$\sim p$	$\sim q$	$p \wedge \sim q$	$\sim p \vee (p \wedge \sim q)$	$(p \wedge q) \vee (\sim p \vee (p \wedge \sim q))$
T	T	T	F	F	F	F	T
T	F	F	F	T	T	T	T
F	T	F	T	F	F	T	T
F	F	F	T	T	F	T	T

Hence $(p \wedge q) \vee (\sim p \vee (p \wedge \sim q)) \equiv t$

EXERCISE:

Use truth table to show that $(p \wedge \sim q) \wedge (\sim p \vee q)$ is a contradiction.

SOLUTION:

Since we have to show that the given statement form is Contradiction so its column in the truth table will have all entries as F.

As clear from the table below

p	q	$\sim q$	$p \wedge \sim q$	$\sim p$	$\sim p \vee q$	$(p \wedge \sim q) \wedge (\sim p \vee q)$
T	T	F	F	F	T	F
T	F	T	T	F	F	F
F	T	F	F	T	T	F
F	F	T	F	T	T	F

USAGE OF "OR" IN ENGLISH

In English language the word **or** is sometimes used in an inclusive sense (p or q or both).

Example: I shall buy a pen or a book.

In the above statement, if you buy a pen or a book in both cases the statement is true and if you buy (both) pen and book then statement is again true. Thus we say in the above statement we use or in inclusive sense.

The word or is sometimes used in an exclusive sense (p or q but not both). As in the below statement

Example: Tomorrow at 9, I'll be in Lahore or Islamabad.

Now in above statement we are using or in exclusive sense because both the statements are true then we have F for the statement.

While defining a disjunction the word or is used in its inclusive sense. Thus the symbol $\vee$ means the "inclusive or"

EXCLUSIVE OR:

When or is used in its exclusive sense, the statement "p or q" means "p or q but not both" or "p or q and not p and q" which translates into symbols as:

$(p \vee q) \wedge \sim (p \wedge q)$

Which is abbreviated as:

$p \oplus q$

or **p XOR q**

TRUTH TABLE FOR EXCLUSIVE OR:

p	q	p∨q	p∧q	~(p∧q)	(p∨q) ∧ ~(p∧q)
T	T	T	T	F	F
T	F	T	F	T	T
F	T	T	F	T	T
F	F	F	F	T	F

Note:
Basically

$$\begin{aligned} p \oplus q &\equiv (p \wedge \sim q) \vee (\sim p \wedge q) \\ &\equiv [p \wedge \sim q) \vee \sim p] \wedge [(p \wedge \sim q) \vee q] \\ &\equiv (p \vee q) \wedge \sim(p \wedge q) \\ &\equiv (p \vee q) \wedge (\sim p \vee \sim q) \end{aligned}$$

Lecture No.3 Laws of Logic

APPLYING LAWS OF LOGIC

Using law of logic, simplify the statement form

$$p \vee [\sim(\sim p \wedge q)]$$

Solution:

$p \vee [\sim(\sim p \wedge q)]$	$\equiv p \vee [\sim(\sim p) \vee (\sim q)]$	DeMorgan's Law
	$\equiv p \vee [p \vee (\sim q)]$	Double Negative Law
	$\equiv [p \vee p] \vee (\sim q)$	Associative Law for $\vee$
	$\equiv p \vee (\sim q)$	Indempotent Law

This is the simplified statement form.

EXAMPLE Using Laws of Logic, verify the logical equivalence

$$\sim(\sim p \wedge q) \wedge (p \vee q) \equiv p$$

$\sim(\sim p \wedge q) \wedge (p \vee q)$	$\equiv (\sim(\sim p) \vee \sim q) \wedge (p \vee q)$	DeMorgan's Law
	$\equiv (p \vee \sim q) \wedge (p \vee q)$	Double Negative Law
	$\equiv p \vee (\sim q \wedge q)$	Distributive Law
	$\equiv p \vee c$	Negation Law
	$\equiv p$	Identity Law

SIMPLIFYING A STATEMENT:

"You will get an A if you are hardworking and the sun shines, or you are hardworking and it rains."

Rephrase the condition more simply.

Solution:

Let p = "You are hardworking'

q = "The sun shines"

r = "It rains" .The condition is then $(p \wedge q) \vee (p \wedge r)$

And using distributive law in reverse,

$$(p \wedge q) \vee (p \wedge r) \equiv p \wedge (q \vee r)$$

Putting $p \wedge (q \vee r)$ back into English, we can rephrase the given sentence as

"You will get an A if you are hardworking and the sun shines or it rains.

EXERCISE:

Use Logical Equivalence to rewrite each of the following sentences more simply.

1. **It is not true that I am tired and you are smart.**
   {I am not tired or you are not smart.}
2. **It is not true that I am tired or you are smart.**
   {I am not tired and you are not smart.}
3. **I forgot my pen or my bag and I forgot my pen or my glasses.**
   {I forgot my pen or I forgot my bag and glasses.
4. **It is raining and I have forgotten my umbrella, or it is raining and I have forgotten my hat.**
   {It is raining and I have forgotten my umbrella or my hat.}

CONDITIONAL STATEMENTS:

Introduction

Consider the statement:

"If you earn an A in Math, then I'll buy you a computer."

This statement is made up of two simpler statements:

p: "You earn an A in Math," and
q: "I will buy you a computer."
The original statement is then saying:
if p is true, then q is true, or, more simply, **if** p, **then** q.
We can also phrase this as p **implies** q, and we write **p → q**.

CONDITIONAL STATEMENTS OR IMPLICATIONS:

If p and q are statement variables, the conditional of q by p is "If p then q" or "p implies q" and is denoted p → q.
It is false when p is true and q is false; otherwise it is true. The arrow "→ " is the **conditional** operator, and in p → q the statement **p** is called the **hypothesis (or antecedent)** and q is called the conclusion (or consequent).

TRUTH TABLE:

p	q	p → q
T	T	T
T	F	F
F	T	T
F	F	T

PRACTICE WITH CONDITIONAL STATEMENTS:

Determine the truth value of each of the following conditional statements:

1. "If 1 = 1, then 3 = 3." **TRUE**
2. "If 1 = 1, then 2 = 3." **FALSE**
3. "If 1 = 0, then 3 = 3." **TRUE**
4. "If 1 = 2, then 2 = 3." **TRUE**
5. "If 1 = 1, then 1 = 2 and 2 = 3." **FALSE**
6. "If 1 = 3 or 1 = 2 then 3 = 3." **TRUE**

ALTERNATIVE WAYS OF EXPRESSING IMPLICATIONS:

The implication **p → q** could be expressed in many alternative ways as:

- "if p then q"
- "p implies q"
- "if p, q"
- "p only if q"
- "p is sufficient for q"
- "not p unless q"
- "q follows from p"
- "q if p"
- "q whenever p"
- "q is necessary for p"

EXERCISE:

Write the following statements in the form "if p, then q" in English.

a) Your guarantee is good only if you bought your CD less than 90 days ago.
If your guarantee is good, then you must have bought your CD player less than 90 days ago.

b) To get tenure as a professor, it is sufficient to be world-famous.
If you are world-famous, then you will get tenure as a professor.

c) That you get the job implies that you have the best credentials.
If you get the job, then you have the best credentials.

d) It is necessary to walk 8 miles to get to the top of the Peak.
If you get to the top of the peak, then you must have walked 8 miles.

TRANSLATING ENGLISH SENTENCES TO SYMBOLS:

Let p and q be propositions:

p = "you get an A on the final exam"
q = "you do every exercise in this book"
r = "you get an A in this class"

Write the following propositions using p, q, and r and logical connectives.

1. To get an A in this class it is necessary for you to get an A on the final.

SOLUTION $p \to r$

2. You do every exercise in this book; You get an A on the final, implies, you get an A in the class.

SOLUTION $p \wedge q \to r$

3. Getting an A on the final and doing every exercise in this book is sufficient for getting an A in this class.

SOLUTION $p \wedge q \to r$

TRANSLATING SYMBOLIC PROPOSITIONS TO ENGLISH:

Let **p**, **q**, and **r** be the propositions:

p = "you have the flu"
q = "you miss the final exam"
r = "you pass the course"

Express the following propositions as an English sentence.

1. $p \to q$

If you have flu, then you will miss the final exam.**2.**$\sim q \to r$

If you don't miss the final exam, you will pass the course.**3.**$\sim p \wedge \sim q \to r$

If you neither have flu nor miss the final exam, then you will pass the course.

HIERARCHY OF OPERATIONS
FOR LOGICAL CONNECTIVES

$\sim$ (negation)

$\wedge$ (conjunction), $\vee$ (disjunction)

$\to$ (conditional)

Construct a truth table for the statement form $p \vee \sim q \to \sim p$

p	q	~q	~p	p ∨ ~q	p ∨ ~ q → ~ p
T	T	F	F	T	F
T	F	T	F	T	F
F	T	F	T	F	T
F	F	T	T	T	T

Construct a truth table for the statement form $(p \to q) \wedge (\sim p \to r)$

p	q	r	$p \to q$	$\sim p$	$\sim p \to r$	$(p \to q) \wedge (\sim p \to r)$
T	T	T	T	F	T	T
T	T	F	T	F	T	T
T	F	T	F	F	T	F
T	F	F	F	F	T	F
F	T	T	T	T	T	T
F	T	F	T	T	F	F
F	F	T	T	T	T	T
F	F	F	T	T	F	F

LOGICAL EQUIVALENCE INVOLVING IMPLICATION

Use truth table to show $p \to q \equiv \sim q \to \sim p$

p	q	$\sim q$	$\sim p$	$p \to q$	$\sim q \to \sim p$
T	T	F	F	T	T
T	F	T	F	F	F
F	T	F	T	T	T
F	F	T	T	T	T

same truth values

Hence the given two expressions are equivalent.

IMPLICATION LAW

$p \to q \equiv \sim p \vee q$

p	q	$p \to q$	$\sim p$	$\sim p \vee q$
T	T	T	F	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

same truth values

NEGATION OF A CONDITIONAL STATEMENT:

Since $p \to q \equiv \sim p \vee q$ therefore

$$\sim(p \to q) \equiv \sim(\sim p \vee q)$$
$$\equiv \sim(\sim p) \wedge (\sim q) \quad \text{by De Morgan's law}$$
$$\equiv p \wedge \sim q \quad \text{by the Double Negative law}$$

Thus the negation of "**if p then q**" is logically equivalent to "**p and not q**".
Accordingly, the negation of an if-then statement does not start with the word if.

EXAMPLES

Write negations of each of the following statements:

1. If Ali lives in Pakistan then he lives in Lahore.
2. If my car is in the repair shop, then I cannot get to class.
3. If x is prime then x is odd **or** x is 2.
4. If n is divisible by 6, then n is divisible by 2 **and** n is divisible by 3.

SOLUTIONS:

1. Ali lives in Pakistan and he does not live in Lahore.
2. My car is in the repair shop and I can get to class.
3. x is prime but x is not odd **and** x is not 2.
4. n is divisible by 6 but n is not divisible by 2 **or** by 3.

INVERSE OF A CONDITIONAL STATEMENT:

The inverse of the conditional statement $\mathbf{p \to q}$ is $\mathbf{\sim p \to \sim q}$

A conditional and its inverse are not equivalent as could be seen from the truth table.

p	q	$p \to q$	$\sim p$	$\sim q$	$\sim p \to \sim q$
T	T	T	F	F	T
T	F	F	F	T	T
F	T	T	T	F	F
F	F	T	T	T	T

different truth values in rows 2 and 3

WRITING INVERSE:

1. ***If today is Friday, then 2 + 3 = 5.***
   If today is not Friday, then $2 + 3 \neq 5$.
2. ***If it snows today, I will ski tomorrow.***
   If it does not snow today I will not ski tomorrow.
3. ***If P is a square, then P is a rectangle.***
   If P is not a square then P is not a rectangle.
4. ***If my car is in the repair shop, then I cannot get to class.***
   If my car is not in the repair shop, then I shall get to the class.

CONVERSE OF A CONDITIONAL STATEMENT:

The converse of the conditional statement $\mathbf{p \to q}$ is $\mathbf{q \to p}$

A conditional and its converse are not equivalent.

i.e., $\to$ is not a commutative operator.

p	q	p→q	q→p
T	T	T	T
T	F	F	T
F	T	T	F
F	F	T	T

not the same

WRITING CONVERSE:

1.If today is Friday, then 2 + 3 = 5.
If 2 + 3 = 5, then today is Friday.

2.If it snows today, I will ski tomorrow.
I will ski tomorrow only if it snows today.

3. If P is a square, then P is a rectangle.
If P is a rectangle then P is a square.

4. If my car is in the repair shop, then I cannot get to class.
If I cannot get to the class, then my car is in the repair shop.

CONTRAPOSITIVE OF A CONDITIONAL STATEMENT:

The contrapositive of the conditional statement $\mathbf{p \to q}$ is $\mathbf{\sim q \to \sim p}$
A conditional and its contrapositive are equivalent. Symbolically$\mathbf{\to q \equiv \sim q \to \sim p}$

1. If today is Friday, then 2 + 3 = 5.
If $2 + 3 \neq 5$, then today is not Friday.

2. If it snows today, I will ski tomorrow.
I will not ski tomorrow only if it does not snow today.

3. If P is a square, then P is a rectangle.
If P is not a rectangle then P is not a square.

4. If my car is in the repair shop, then I cannot get to class.
If I get to the class, then my car is not in the repair shop.

Lecture No.4 Biconditional

BICONDITIONAL

If p and q are statement variables, the biconditional of p and q is
"p if, and only if, q" and is denoted **$p \leftrightarrow q$**. *if and only if* abbreviated **iff.**
The double headed arrow " $\leftrightarrow$" is the **biconditional operator.**
TRUTH TABLE FOR
$p \leftrightarrow q$.

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

EXAMPLES:

True or false?

1. "**1+1 = 3** if and only if **earth is flat**"
   TRUE
2. "**Sky is blue** iff **1 = 0**"
   FALSE3. "Milk is white iff **birds lay eggs"**
   TRUE
4. "**33 is divisible by 4** if and only if **horse has four legs**"
   FALSE
5. "**$x > 5$ iff $x^2 > 25$**"
   FALSE

$p \leftrightarrow q \equiv (p \to q) \wedge (q \to p)$

p	q	$p \leftrightarrow q$	$p \to q$	$q \to p$	$(p \to q) \wedge (q \to p)$
T	T	T	T	T	T
T	F	F	F	T	F
F	T	F	T	F	F
F	F	T	T	T	T

same truth values

REPHRASING BICONDITIONAL:

$p \leftrightarrow q$ is also expressed as:
"p is necessary and sufficient for q"

"if p then q, and conversely"
"p is equivalent to q"

EXERCISE:

Rephrase the following propositions in the form "p if and only if q" in English.

1. If it is hot outside you buy an ice cream cone, and if you buy an ice cream cone it is hot outside.

Sol You buy an ice cream cone if and only if it is hot outside.

2. For you to win the contest it is necessary and sufficient that you have the only winning ticket.

Sol You win the contest if and only if you hold the only winning ticket.

3. If you read the news paper every day, you will be informed and conversely.

Sol You will be informed if and only if you read the news paper every day.**4.It rains if it is a weekend day, and it is a weekend day if it rains.**

Sol It rains if and only if it is a weekend day.

5. The train runs late on exactly those days when I take it.

Sol The train runs late if and only if it is a day I take the train.

6. This number is divisible by 6 precisely when it is divisible by both 2 and 3.

Sol This number is divisible by 6 if and only if it is divisible by both 2 and 3.

TRUTH TABLE FOR
$(p\to q)\leftrightarrow(\sim q\to\sim p)$

p	q	$p\to q$	$\sim q$	$\sim p$	$\sim q\to\sim p$	$(p\to q)\leftrightarrow(\sim q\to\sim p)$
T	T	T	F	F	T	T
T	F	F	T	F	F	T
F	T	T	F	T	T	T
F	F	T	T	T	T	T

TRUTH TABLE FOR
$(p\leftrightarrow q)\leftrightarrow(r\leftrightarrow q)$

p	q	r	$p\leftrightarrow q$	$r\leftrightarrow q$	$(p\leftrightarrow q)\leftrightarrow(r\leftrightarrow q)$
T	T	T	T	T	T
T	T	F	T	F	F
T	F	T	F	F	T
T	F	F	F	T	F
F	T	T	F	T	F
F	T	F	F	F	T
F	F	T	T	F	F
F	F	F	T	T	T

TRUTH TABLE FOR

$p \wedge \sim r \leftrightarrow q \vee r$

Here $p \wedge \sim r \leftrightarrow q \vee r$ means $(p \wedge (\sim r)) \leftrightarrow (q \vee r)$

p	q	r	~r	p∧~r	q∨r	p ∧ ~r ↔q∨r
T	T	T	F	F	T	F
T	T	F	T	T	T	T
T	F	T	F	F	T	F
T	F	F	T	T	F	F
F	T	T	F	F	T	F
F	T	F	T	F	T	F
F	F	T	F	F	T	F
F	F	F	T	F	F	T

LOGICAL EQUIVALENCE

INVOLVING BICONDITIONAL

Show that **$\sim p \leftrightarrow q$** and **$p \leftrightarrow \sim q$** are logically equivalent

p	q	~p	~q	~p↔q	p↔~q
T	T	F	F	F	F
T	F	F	T	T	T
F	T	T	F	T	T
F	F	T	T	F	F

same truth values

EXERCISE:

Show that **~(p⊕q)** and **p↔q** are logically equivalent

p	q	p⊕q	~(p⊕q)	p↔q
T	T	F	T	T
T	F	T	F	F
F	T	T	F	F
F	F	F	T	T

same truth values

LAWS OF LOGIC:

1. Commutative Law: $p \leftrightarrow q \equiv q \leftrightarrow p$
2. Implication Laws: $p \to q \equiv \sim p \vee q$
   $\equiv \sim(p \wedge \sim q)$
3. Exportation Law: $(p \wedge q) \to r \equiv p \to (q \to r)$
4. Equivalence: $p \leftrightarrow q \equiv (p \to q) \wedge (q \to p)$
5. Reductio ad absurdum $p \to q \equiv (p \wedge \sim q) \to c$

APPLICATION:

Rewrite the statement forms without using the symbols → or ↔

1. $p \wedge \sim q \to r$ **2. $(p \to r) \leftrightarrow (q \to r)$**

SOLUTION

1. $p \wedge \sim q \to r \equiv (p \wedge \sim q) \to r$ order of operations
   $\equiv \sim(p \wedge \sim q) \vee r$ implication law
2. $(p \to r) \leftrightarrow (q \to r) \equiv (\sim p \vee r) \leftrightarrow (\sim q \vee r)$ implication law
   $\equiv [(\sim p \vee r) \to (\sim q \vee r)] \wedge [(\sim q \vee r) \to (\sim p \vee r)]$
   equivalence of biconditional
   $\equiv [\sim(\sim p \vee r) \vee (\sim q \vee r)] \wedge [\sim(\sim q \vee r) \vee (\sim p \vee r)]$
   implication law

Rewrite the statement form **$\sim p \vee q \to r \vee \sim q$** to a logically equivalent form that uses only ~ and ∧

SOLUTION

STATEMENT	REASON
$\sim p \vee q \to r \vee \sim q$	Given statement form
$\equiv (\sim p \vee q) \to (r \vee \sim q)$	Order of operations
$\equiv \sim[(\sim p \vee q) \wedge \sim (r \vee \sim q)]$	Implication law $p \to q \equiv \sim(p \wedge \sim q)$
$\equiv \sim[\sim(p \wedge \sim q) \wedge (\sim r \wedge q)]$	De Morgan's law

Show that **$\sim(p \to q) \to p$** is a tautology without using truth tables.

SOLUTION

STATEMENT	REASON
$\sim(p \to q) \to p$	Given statement form
$\equiv \sim[\sim(p \wedge \sim q)] \to p$	Implication law $p \to q \equiv \sim(p \wedge \sim q)$
$\equiv (p \wedge \sim q) \to p$	Double negation law
$\equiv \sim(p \wedge \sim q) \vee p$	Implication law $p \to q \equiv \sim p \vee q$
$\equiv (\sim p \vee q) \vee p$	De Morgan's law
$\equiv (q \vee \sim p) \vee p$	Commutative law of
$\vee\equiv q \vee (\sim p \vee p)$	Associative law of
$\vee\equiv q \vee t$	Negation law
$\equiv t$	Universal bound law

EXERCISE:

Suppose that p and q are statements so that $p \to q$ is false. Find the truth values of each of the following:

1. $\sim p \to q$
2. $p \vee q$
3. $q \leftrightarrow p$

SOLUTION

1. TRUE
2. TRUE
3. FALSE

Lecture No.9 Set identities

SET IDENTITIES:

Let A, B, C be subsets of a universal set U.

1. Idempotent Laws
   a. $A \cup A = A$ b. $A \cap A = A$
2. Commutative Laws
   a. $A \cup B = B \cup A$ b. $A \cap B = B \cap A$
3. Associative Laws
   a. $A \cup (B \cup C) = (A \cup B) \cup C$
   b. $A \cap (B \cap C) = (A \cap B) \cap C$
4. Distributive Laws
   a. $A \cup (B \cap C) = (A \cup B) \cap (A \cup B)$
   b. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
5. Identity Laws
   a. $A \cup \varnothing = A$ b. $A \cap \varnothing = \varnothing$
   c. $A \cup U = U$ d. $A \cap U = A$
6. Complement Laws
   a. $A \cup A^c = U$ b. $A \cap A^c = \varnothing$
   c. $U^c = \varnothing$ d. $\varnothing^c = U$
8. Double Complement Law
   $(A^c)^c = A$
9. DeMorgan's Laws
   a. $(A \cup B)^c = A^c \cap B^c$ b. $(A \cap B)^c = A^c \cup B^c$
10. Alternative Representation for Set Difference
   $A - B = A \cap B^c$
11. Subset Laws
   a. $A \cup B \subseteq C$ *iff* $A \subseteq C$ and $B \subseteq C$
   b. $C \subseteq A \cap B$ *iff* $C \subseteq A$ and $C \subseteq B$
12. Absorption Laws
   a. $A \cup (A \cap B) = A$ b. $A \cap (A \cup B) = A$

EXERCISE:

1. **$A \subseteq A \cup B$**
2. **$A - B \subseteq A$**
3. **If $A \subseteq B$ and $B \subseteq C$ then $A \subseteq C$**
4. **$A \subseteq B$ if, and only if, $B^c \subseteq A^c$**

1. **Prove that $A \subseteq A \cup B$**

SOLUTION

Here in order to prove the identity you should remember the definition of Subset of a set. We will take the arbitrary element of a set then show that, that element is the member of the other then the first set is the subset of the other. So

Let x be an arbitrary element of A, that is $x \in A$.

$\Rightarrow$ $x \in A$ or $x \in B$

$\Rightarrow \quad x \in A \cup B$

But x is an arbitrary element of A.

$\therefore \quad A \subseteq A \cup B \qquad$ (proved)

1. **Prove that $A - B \subseteq A$**

SOLUTION

Let $x \in A - B$

$\Rightarrow \quad x \in A$ and $x \notin B \qquad$ (by definition of $A - B$)

$\Rightarrow \quad x \in A \qquad$ (in particular)

But x is an arbitrary element of $A - B$

$\therefore \quad A - B \subseteq A \qquad$ (proved)

1. **Prove that if $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$**

SOLUTION

Suppose that $A \subseteq B$ and $B \subseteq C$

Consider $x \in A$

$\Rightarrow \quad x \in B \qquad$ (as $A \subseteq B$)

$\Rightarrow \quad x \in C \qquad$ (as $B \subseteq C$)

But x is an arbitrary element of A

$\therefore \quad A \subseteq C \qquad$ (proved)

1. **Prove that $A \subseteq B$ iff $B^c \subseteq A^c$**

SOLUTION:

Suppose $A \subseteq B \qquad$ {To prove $B^c \subseteq A^c$}

Let $x \in B^c$

$\Rightarrow \quad x \notin B \qquad$ (by definition of B^c)

$\Rightarrow \quad x \notin A$

$\Rightarrow \quad x \in A^c \qquad$ (by definition of A^c)

Now we know that implication and its contrapositivity are logically equivalent and the contrapositive statement of if $x \in A$ then $x \in B$ is: if $x \notin B$ then $x \notin A$ which is the definition of the $A \subseteq B$. Thus if we show for any two sets A and B, if $x \notin B$ then $x \notin A$ it means that

$A \subseteq B$. Hence

But x is an arbitrary element of B^c

$\therefore B^c \subseteq A^c$

Conversely,

Suppose $B^c \subseteq A^c \qquad$ {To prove $A \subseteq B$}

Let $x \in A$

$\Rightarrow \quad x \notin A^c \qquad$ (by definition of A^c)

$\Rightarrow \quad x \notin B^c \qquad$ ($\therefore B^c \subseteq A^c$)

$\Rightarrow \quad x \in B \qquad$ (by definition of B^c)

But x is an arbitrary element of A.

$\therefore A \subseteq B \qquad$ (proved)

EXERCISE:

Let A and B be subsets of a universal set U.

Prove that $A - B = A \cap B^c$.

SOLUTION

Let $x \in A - B$

$\Rightarrow x \in A$ and $x \notin B$ (definition of set difference)

$\Rightarrow x \in A$ and $x \in B^c$ (definition of complement)

$\Rightarrow x \in A \cap B^c$ (definition of intersection)

But x is an arbitrary element of $A - B$ so we can write

$\therefore A - B \subseteq A \cap B^c$.............(1)

Conversely,

let $y \in A \cap B^c$

$\Rightarrow y \in A$ and $y \in B^c$ (definition of intersection)

$\Rightarrow y \in A$ and $y \notin B$ (definition of complement)

$\Rightarrow y \in A - B$ (definition of set difference)

But y is an arbitrary element of $A \cap B^c$

$\therefore A \cap B^c \subseteq A - B$.............(2)

From (1) and (2) it follows that

$A - B = A \cap B^c$ (as required)

EXERCISE:

Prove the DeMorgan's Law: $(A \cup B)^c = A^c \cap B^c$

PROOF

Let $x \in (A \cup B)^c$

$\Rightarrow x \notin A \cup B$ (definition of complement)

$x \notin A$ and $x \notin B$ (DeMorgan's Law of Logic)

$\Rightarrow x \in A^c$ and $x \in B^c$ (definition of complement)

$\Rightarrow x \in A^c \cap B^c$ (definition of intersection)

But x is an **arbitrary** element of $(A \cup B)^c$ so we have proved that

$\therefore (A \cup B)^c \subseteq A^c \cap B^c$.........(1)

Conversely

let $y \in Ac \cap B^c$

$\Rightarrow y \in A^c$ and $y \in B^c$ (definition of intersection)

$\Rightarrow y \notin A$ and $y \notin B$ (definition of complement)

$\Rightarrow y \notin A \cup B$ (DeMorgan's Law of Logic)

$\Rightarrow y \in (A \cup B)^c$ (definition of complement)

But y is an arbitrary element of $A^c \cap B^c$

$\therefore A^c \cap B^c \subseteq (A \cup B)^c$..................(2)

From (1) and (2) we have

$(A \cup B)^c = A^c \cap B^c$

Which is the Demorgan's Law.

EXERCISE:

Prove the associative law: $A \cap (B \cap C) = (A \cap B) \cap C$

PROOF:

Consider $x \in A \cap (B \cap C)$

$\Rightarrow x \in A$ and $x \in B \cap C$ (definition of intersection)

$\Rightarrow x \in A$ and $x \in B$ and $x \in C$ (definition of intersection)

$\Rightarrow x \in A \cap B$ and $x \in C$ (definition of intersection)

$\Rightarrow x \in (A \cap B) \cap C$ (definition of intersection)

But x is an arbitrary element of $A \cap (B \cap C)$

$$\therefore A \cap (B \cap C) \subseteq (A \cap B) \cap C \ldots\ldots(1)$$

Conversely

let $y \in (A \cap B) \cap C$

$\Rightarrow y \in A \cap B$ and $y \in C$ (definition of intersection)

$\Rightarrow y \in A$ and $y \in B$ and $y \in C$ (definition of intersection)

$\Rightarrow y \in A$ and $y \in B \cap C$ (definition of intersection)

$\Rightarrow y \in A \cap (B \cap C)$ (definition of intersection)

But y is an arbitrary element of $(A \cap B) \cap C$

$$\therefore (A \cap B) \cap C \subseteq A \cap (B \cap C) \ldots\ldots..(2)$$

From (1) & (2), we conclude that

$A \cap (B \cap C) = (A \cap B) \cap C$ (proved)

EXERCISE:

Prove the distributive law: $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

PROOF:

Let $x \in A \cup (B \cap C)$

$\Rightarrow x \in A$ or $x \in B \cap C$ (definition of union)

Now since we have $x \in A$ or $x \in B \cap C$ it means that either x is in A or in $A \cap B$ it is in the $A \cup (B \cap C)$ so in order to show that $A \cup (B \cap C)$ is the subset of $(A \cup B) \cap (A \cup C)$ we will consider both the cases when x is iu A or x is in $B \cap C$ hence we will consider the two cases.

CASE I:

(when $x \in A$)

$\Rightarrow x \in A \cup B$ and $x \in A \cup C$ (definition of union)

Hence,

$x \in (A \cup B) \cap (A \cup C)$ (definition of intersection)

CASE II:

(when $x \in B \cap C$)

We have $x \in B$ and $x \in C$ (definition of intersection)

Now $x \in B \Rightarrow x \in A \cup B$ (definition of union)

and $x \in C \Rightarrow x \in A \cup C$ (definition of union)

Thus $x \in A \cup B$ and $x \in A \cup C$

$\Rightarrow x \in (A \cup B) \cap (A \cup C)$

In both of the cases $x \in (A \cup B) \cap (A \cup C)$

Accordingly,

$$A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C) \ldots\ldots..(1)$$

Conversely,

Suppose $x \in (A \cup B) \cap (A \cup C)$

$\Rightarrow x \in (A \cup B)$ and $x \in (A \cup C)$ (definition of intersection)

Consider the two cases $x \in A$ and $x \notin A$

CASE I: (when $x \in A$)

We have $x \in A \cup (B \cap C)$ (definition of union)

CASE II: (when $x \notin A$)

Since $x \in A \cup B$ and $x \notin A$, therefore $x \in B$

Also, since $x \in A \cup C$ and $x \notin A$, therefore $x \in C$. Thus $x \in B$ and $x \in C$
That is, $x \in B \cap C$

$\Rightarrow x \in A \cup (B \cap C)$ (definition of union)

Hence in both cases
$x \in A \cup (B \cap C)$

$\therefore (A \cup B) \cap C\ (A \cup C) \subseteq A \cup (B \cap C)$........(2)

By (1) and (2), it follows that
$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ (proved)

EXERCISE:

For any sets A and B if $A \subseteq B$ then
(a) $A \cap B = A$ (b) $A \cup B = B$

SOLUTION:

(a) Let $x \in A \cap B$
$\Rightarrow x \in A$ and $x \in B$
$\Rightarrow x \in A$ (in particular)
Hence $A \cap B \subseteq A$..............(1)

Conversely,

let $x \in A$.
Then $x \in B$ (since $A \subseteq B$)
Now $x \in A$ and $x \in B$, therefore $x \in A \cap B$
Hence, $A \subseteq A \cap B$..............(2)
From (1) and (2) it follows that
$A = A \cap B$ (proved)

(b) Prove that $A \cup B = B$ when $A \subseteq B$

SOLUTION:

Suppose that $A \subseteq B$. Consider $x \in A \cup B$.

CASE I (when $x \in A$)
Since $A \subseteq B$, $x \in A \Rightarrow x \in B$

CASE II (when $x \notin A$)
Since $x \in A \cup B$, we have $x \in B$

Thus $x \in B$ in both the cases, and we have
$A \cup B \subseteq B$...............(1)

Conversely

let $x \in B$. Then clearly, $x \in A \cup B$
Hence $B \subseteq A \cup B$................(2)
Combining (1) and (2), we deduce that
$A \cup B = B$ (proved)

USING SET IDENTITIES:

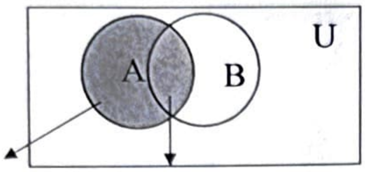
For all subsets A and B of a universal set U, prove that
$$(A - B) \cup (A \cap B) = A$$

PROOF:

LHS $= (A - B) \cup (A \cap B)$
$= (A \cap B^c) \cup (A \cap B)$ (Alternative representation for set difference)

$= A \cap (B^c \cup B)$ Distributive Law
$= A \cap U$ Complement Law
$= A$ Identity Law
$= RHS$ (proved)

The result can also be seen by Venn diagram.



A-B $A \cap B$

EXERCISE:

For any two sets A and B prove that $A - (A - B) = A \cap B$

SOLUTION

LHS $= A - (A - B)$
$= A - (A \cap B^c)$ Alternative representation for set difference
$= A \cap (A \cap B^c)^c$ Alternative representation for set difference
$= A \cap (A^c \cup (B^c)^c)$ DeMorgan's Law
$= A \cap (A^c \cup B)$ Double Complement Law
$= (A \cap A^c) \cup (A \cap B)$ Distributive Law
$= \varnothing \cup (A \cap B)$ Complement Law
$= A \cap B$ Identity Law
$= RHS$ (proved)

EXERCISE:

For all set A, B, and C prove that $(A - B) - C = (A - C) - B$

SOLUTION

LHS $= (A - B) - C$
$= (A \cap B^c) - C$ Alternative representation of set difference
$= (A \cap B^c) \cap C^c$ Alternative representation of set difference
$= A \cap (B^c \cap C^c)$ Associative Law
$= A \cap (C^c \cap B^c)$ Commutative Law
$= (A \cap C^c) \cap B^c$ Associative Law
$= (A - C) \cap B^c$ Alternative representation of set difference
$= (A - C) - B$ Alternative representation of set difference
$= RHS$ (proved)

EXERCISE:

Simplify $(B^c \cup (B^c - A))^c$

SOLUTION

$(B^c \cup (B^c - A))^c = (B^c \cup (B^c \cap A^c))^c$
Alternative representation for set difference
$= (B^c)^c \cap (B^c \cap A^c)^c$ DeMorgan's Law
$= B \cap ((B^c)^c \cup (A^c)^c)$ DeMorgan's Law
$= B \cap (B \cup A)$ Double Complement Law

$= B$ Absorption Law

is the simplified form of the given expression.

PROVING SET IDENTITIES BY MEMBERSHIP TABLE:

Prove the following using Membership Table:

(i) $A - (A - B) = A \cap B$

(ii) $(A \cap B)^c = A^c \cup B^c$

(iii) $A - B = A \cap B^c$

$\mathbf{A - (A - B) = A \cap B}$

A	B	A-B	A-(A-B)	A∩B
1	1	0	1	1
1	0	1	0	0
0	1	0	0	0
0	0	0	0	0

$\mathbf{(A \cap B)^c = A^c \cup B^c}$

A	B	$A \cap B$	$(A \cap B)^c$	A^c	B^c	$A^c \cup B^c$
1	1	1	0	0	0	0
1	0	0	1	0	1	1
0	1	0	1	1	0	1
0	0	0	1	1	1	1

SOLUTION (iii):

A	B	$A - B$	B^c	$A \cap B^c$
1	1	0	0	0
1	0	1	1	1
0	1	0	0	0
0	0	0	1	0

Lecture No.10 Applications of Venn diagram

Exercise:

A number of computer users are surveyed to find out if they have a printer, modem or scanner. Draw separate Venn diagrams and shade the areas, which represent the following configurations.

(i) modem and printer but no scanner
(ii) scanner but no printer and no modem
(iii) scanner or printer but no modem.
(iv) no modem and no printer.

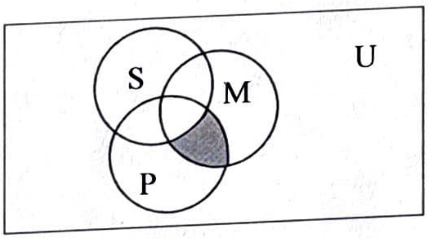
SOLUTION

Let

P represent the set of computer users having printer.
M represent the set of computer users having modem.
S represent the set of computer users having scanner.

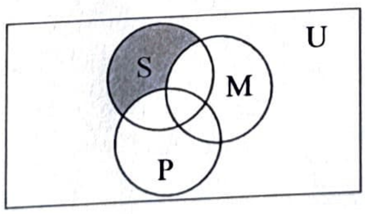
SOLUTION (i)

Modem and printer but no Scanner is shaded.



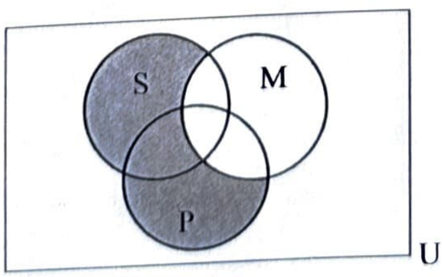
SOLUTION (ii)

Scanner but no printer and no modem is shaded.



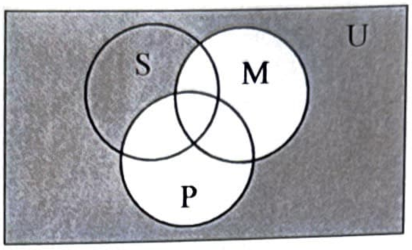
SOLUTION (iii)

scanner or printer but no modem is shaded.



SOLUTION (iv)

no modem and no printer is shaded.



EXERCISE:

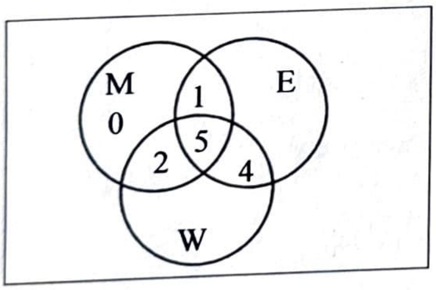
Of **21 typists** in an office, **5** use all **manual typewriters (M), electronic typewriters (E) and word processors (W); 9** use **E** and **W; 7** use **M** and **W; 6** use **M** and **E;** but **no one** uses **M** only.

(i) Represent this information in a Venn Diagram.

(ii) If the same number of typists use electronic as use word processors, then

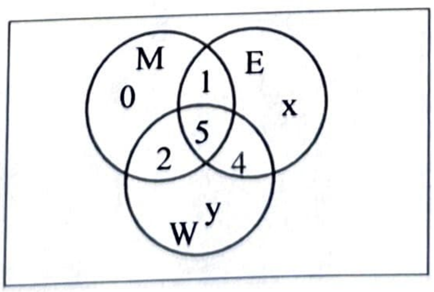
1. (a) How many use word processors only,
2. (b) How many use electronic typewriters?

SOLUTION (i)



SOLUTION (ii-a)

Let the number of typists using electronic typewriters (E) only be x, and the number of typists using word processors (W) only be y.



Total number of typists using E = Total Number of typists using W

$$1 + 5 + 4 + x = 2 + 5 + 4 + y$$

$$\text{or,} \qquad x - y = 1 \qquad \ldots\ldots\ldots\ldots(1)$$

Also, total number of typists = 21

$$\Rightarrow \quad 0 + x + y + 1 + 2 + 4 + 5 = 21$$

$$\text{or,} \qquad x + y = 9 \qquad \ldots\ldots\ldots\ldots(2)$$

Solving (1) & (2), we get

$$x = 5, \quad y = 4$$

$\therefore$ Number of typists using word processor only is $y = 4$

(ii)-(b) How many typists use electronic typewriters?

SOLUTION:

Typists using electronic typewriters = No. of elements in E

$$= 1 + 5 + 4 + x$$
$$= 1 + 5 + 4 + 5$$
$$= 15$$

EXERCISE

In a school, 100 students have access to three software packages, A, B and C

28 did not use any software

8 used only packages A

26 used only packages B

7 used only packages C

10 used all three packages

13 used both A and B

(i) Draw a Venn diagram with all sets enumerated as for as possible. Label the two subsets which cannot be enumerated as x and y, in any order.

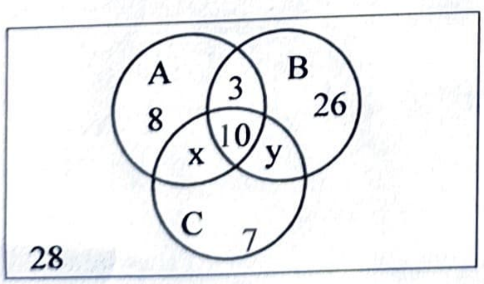
(ii) If twice as many students used package B as package A, write down a pair of simultaneous equations in x and y.

(iii) Solve these equations to find x and y.

(iv) How many students used package C?

SOLUTION(i)

Venn Diagram with all sets enumerated.



(ii) If twice as many students used package B as package A, write down a pair of simultaneous equations in x and y.

SOLUTION:

We are given

students using package B = 2 (# students using package A)

Now the number of students which used package B and A are clear from the diagrams given below. So we have the following equation

$$\Rightarrow \quad 3 + 10 + 26 + y = 2(8 + 3 + 10 + x)$$
$$\Rightarrow \quad 39 + y = 42 + 2x$$
$$\text{or} \quad y = 2x + 3 \quad \ldots\ldots\ldots\ldots(1)$$

Also, total number of students = 100.

Hence, $8 + 3 + 26 + 10 + 7 + 28 + x + y = 100$

or $82 + x + y = 100$ …………(2)

or $x + y = 18$

(iii)Solving simultaneous equations for x and y.

SOLUTION:

$y = 2x + 3$ ………………(1)

$x + y = 18$ ………………(2)

Using (1) in (2), we get,

$$x + (2x + 3) = 18$$
$$\text{or} \quad 3x + 3 = 18$$
$$\text{or} \quad 3x = 15$$
$$\Rightarrow x = 5$$

Consequently $y = 13$

How many students used package C?

SOLUTION:

No. of students using package C

$$= x + y + 10 + 7$$
$$= 5 + 13 + 10 + 7$$
$$= 35$$

EXAMPLE:

Use diagrams to show the validity of the following argument:

All human beings are mortal

Zeus is not mortal

$\therefore$ Zeus is not a human being

SOLUTION:

The premise "All human beings are mortal is pictured by placing a disk labeled "human beings" inside a disk labeled "mortals". We place the disk of human Beings in side the Disk of mortals because there are things which are mortal but not Human beings so the set of human beings is subset of set of Mortals.